A-TO-Z COMPLETE TECHNICAL SPECIFICATION REPORT

Commercial AI Physiotherapy & Rehabilitation Platform

Document Version: 3.2.0 (Stage 1 Production Ready)	**Date:** August 2026
Target Domain: AI Physiotherapy & Goniometry	**System FPS:** 34.6 - 35.1 FPS CPU Multi-Threaded

1. Executive Overview & Commercial Scope

The **Commercial AI Physiotherapy Platform** is a high-precision, real-time medical computer vision system designed for clinical rehabilitation, joint Range of Motion (ROM) assessment, posture evaluation, and movement analysis. The engine processes live 2D/3D webcam feeds to compute 23 clinical goniometric joint angles, track 21 finger joints per hand, and generate dynamic natural language feedback instructions without requiring specialized hardware or wearable sensors.

2. Technology Stack & Installed Libraries

Component / Library	Version	Role & Clinical Functionality
Python Core	3.10+	Primary runtime environment
MediaPipe Pose	0.10.x	33-Landmark 3D Human Pose Estimation
MediaPipe Hands	0.10.x	21-Landmark Hand & Finger Joint Tracking
OpenCV (opencv-python)	4.8.x	Image processing, anti-aliased skeleton rendering, HUD UI
NumPy	1.24+	Vector mathematics, matrix operations, angle calculation
SciPy	1.11+	Signal filtering, spatial algorithms, statistical metrics
ONNX Runtime	1.16+	CPU multi-threaded inference acceleration
PyYAML	6.0+	Configuration management & YAML database parser
ReportLab	4.0+	Clinical session report PDF generation engine

3. 3-Thread Parallel System Architecture

To guarantee real-time execution (>30 FPS) on standard CPU hardware, the application utilizes a **3-Thread Parallel Pipeline** with thread-safe queue communication:

Thread Name	Module File	Responsibilities & Operation
Thread 1: Camera Producer	camera/stream.py	Asynchronous webcam capture at 60 FPS, frame buffer queue managem
Thread 2: Inference Worker	inference/pipeline.py	MediaPipe 3D Landmark extraction, One-Euro temporal filtering, goniom
Thread 3: Renderer Main Thread	visualization/medical_gui.py	Anti-aliased stick figure skeleton rendering, HUD overlays, goniometer ar

4. Clinical Biomechanical Goniometry Equations

A. Anatomical Torso Vertical Axis Abduction Vector:
Torso Vertical Reference Axis: **v_torso = hip_center - shoulder_center**
Arm Vector: **v_arm = p_elbow - p_shoulder**
Abduction Angle: **theta_abduction = arccos((v_arm · v_torso) / (||v_arm|| ||v_torso||))**
• Arm at side = 0°, Arm horizontal = 90°, Arm overhead = 180°.

B. Standard 3-Point Goniometric Elbow & Knee Flexion:
Elbow Flexion = **|180° - angle(Shoulder, Elbow, Wrist)|**
Knee Flexion = **|180° - angle(Hip, Knee, Ankle)|**
Ankle Angle = **angle(Knee, Ankle, FootIndex)** (90° = plantigrade neutral foot).

C. Front-View vs Sagittal-View Flexion Handling:
True Shoulder Flexion is a sagittal plane (side-view) movement. Front-facing cameras display **'SIDE VIEW REQ'** for Shoulder Flexion and present **Shoulder Abduction** as the primary frontal plane metric.

5. Selective Body Tracking Modes (Upper, Lower, Full)

The system supports 3 selective anatomical tracking modes chosen on launch or toggled dynamically:

Mode Name	Target Landmarks	HUD Displayed Cards & Features
UPPER_BODY	Nose, Neck, Shoulders, Elbows, Wrists, Hands, Spine	Left & Right Upper Body Cards (Elbow Flexion, Shoulder Abduc
LOWER_BODY	Pelvis, Hips, Knees, Ankles, Heels, Feet	Left & Right Lower Body Cards (Hip Flexion, Knee Flexion, Ank
FULL_BODY	All 33 Full-Body Landmarks + 42 Hand Landmarks	Complete Full Body Skeleton, all HUD Cards, Movement Qualit

6. Complete File Tree & Module Directory Guide

File / Directory Path	Component Description & Primary Role
live_pose.py	Main Application Entry Point. Initializes 3-thread parallel camera pipeline, interactive star
metrics/physio_angles.py	Clinical Goniometric Angle Calculation Engine. Computes 3D Torso Vertical Abduction, E
src/physio_analysis.py	Physiotherapy Analysis Engine. Computes real-time Movement Quality Score %, Range
src/physio_reporter.py	Clinical Session Reporter. Exports clinical reports to JSON and Markdown artifacts.
src/camera_validator.py	Selective Camera Positioning & Framing Engine. Validates landmark visibility for UPPER
visualization/medical_gui.py	Commercial Medical GUI Renderer. Draws anti-aliased stick figure skeleton, 21-joint fing
camera/stream.py	Asynchronous Threaded Webcam Stream Class.
inference/pipeline.py	Real-Time Inference Pipeline wrapper for MediaPipe Pose & Hands.
tracking/tracker.py	Multi-Person IoU & Centroid Joint Tracker.
utilities/logger.py	Session Video Recorder & Screenshot Logger.
validate_clinical_angles.py	Clinical Goniometric Angle Audit Suite (Validates MAE across 73 real human subject pho
test_shoulder_abduction_validation.py	Standalone 6-Pose Mathematical Validation Test Suite for Shoulder Abduction.
test_physio_session.py	System Integration Test Suite.
benchmark.py	System Performance & Accuracy Benchmark.
config/config.yaml	System Configuration Parameters.

7. Complete Execution Command Reference Guide

Command	Description & Expected Output
python live_pose.py	Launch Live AI Physiotherapy WebCam Application. Displays interactive startup me
python test_shoulder_abduction_validation.py	Run Standalone 6-Pose Shoulder Abduction Validation Test Suite. Outputs mathem
python validate_clinical_angles.py	Run Clinical Goniometric Angle Audit across 73 real human photos. Outputs per-joir
python test_physio_session.py	Run Integration Test Suite. Computes ROM summary, renders annotated GUI image
python benchmark.py	Run System Performance & Accuracy Benchmark. Benchmarks 100 frames and exp

8. Interactive Keyboard Hotkey Quick Reference

Hotkey Key	Action & System Response
1 or U	Switch Body Mode to UPPER BODY MODE
2 or L	Switch Body Mode to LOWER BODY MODE
3 or F	Switch Body Mode to FULL BODY MODE
D	Toggle Debug Mode HUD (Displays abduction angles, torso tilt, landmark confidence)
R	Toggle MP4 Video Session Recording
S	Capture High-Resolution PNG Screenshot
H	Toggle 21-Joint Finger & Hand Skeleton Overlay
C	Toggle Person Confidence & Tracking Badges
A	Toggle Translucent Angle HUD Cards
B	Toggle Skeleton Bone Links
J	Toggle Glowing Joint Nodes
Q or Esc	Clean Application Exit & Resource Release

End of Official Technical Report — Full Body AI Medical Physiotherapy Platform